

Towards Robust Music Genre Classification: A Comparative Study on the GTZAN Dataset

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Abstract— Music genre classification plays an important role in modern music recommendation and streaming systems by enabling automatic categorization of audio tracks. Manual genre labeling is often inconsistent and time-consuming, motivating the need for intelligent automated classification techniques. This study presents a comparative analysis of multiple machine learning and deep learning approaches for music genre classification using the GTZAN dataset. Traditional machine learning models were trained using handcrafted audio features extracted from both 30-second and 3-second audio clips. Principal Component Analysis (PCA) was further applied to study the impact of dimensionality reduction on classification performance. In addition, a spectrogram-based 2D Convolutional Neural Network (CNN) was implemented to learn visual audio representations from spectrogram images. The study also explores pretrained audio representation models for extracting rich music embeddings, which were combined with Attention and Bidirectional Gated Recurrent Unit (BiGRU) architectures for genre prediction. Experimental evaluation was performed using accuracy, macro F1-score, and confusion matrices. Comparative analysis shows that pretrained embedding-based deep learning approaches outperform traditional handcrafted feature-based methods by capturing more discriminative audio characteristics. The study highlights the effectiveness of modern representation learning techniques for automated music genre recognition.

Keywords— *Deep Learning, Convolutional Neural Network (CNN), Principal Component Analysis (PCA), Audio Signal Processing, Spectrogram Analysis, Pretrained Audio Representation, Attention Mechanism, BiGRU, Music Information Retrieval.*

I. INTRODUCTION

The rapid expansion of digital music platforms and online streaming services has significantly increased the amount of audio content generated and shared across the internet. This growth has created a strong demand for intelligent music organization and retrieval systems capable of automatically categorizing music tracks into predefined genres. Music genre classification is an important research area in Music Information Retrieval, where audio signals are analysed to identify musical styles such as classical, jazz, blues, hip-hop, pop, rock, reggae, disco, and country. Accurate genre recognition plays a major role in recommendation systems, playlist generation, audio indexing, personalized streaming, and music archive management. However, manual genre labelling is often time-consuming, inconsistent, and subjective due to overlapping musical characteristics between genres.

To address these limitations, machine learning and deep learning techniques have been widely explored for automated music genre recognition. Earlier approaches primarily relied on handcrafted audio features extracted from music signals using signal processing techniques. Features such as Mel

Frequency Cepstral Coefficients (MFCCs), chroma features, spectral centroid, spectral bandwidth, roll-off frequency, and zero-crossing rate were commonly used to represent the acoustic properties of music tracks. Traditional machine learning algorithms including Logistic Regression, Support Vector Machines (SVM), Random Forest, and Gradient Boosting classifiers were then applied to classify genres based on these engineered features. Although these approaches achieved reasonable performance, their effectiveness heavily depended on the quality of handcrafted feature extraction and their ability to capture discriminative audio patterns.

Recent advancements in deep learning have significantly improved audio classification systems by enabling models to automatically learn hierarchical feature representations directly from audio data. Convolutional Neural Networks (CNNs) have demonstrated strong performance in spectrogram-based music classification tasks by learning spatial and temporal audio patterns from visual spectrogram representations. Spectrogram images provide a time-frequency representation of audio signals, allowing deep neural networks to identify complex musical structures and frequency variations more effectively than traditional handcrafted descriptors. Despite these improvements, deep learning models often require large datasets and high computational resources for efficient training. In addition to spectrogram-based learning, pretrained audio representation models have recently gained significant attention in music understanding tasks. These models are trained on large-scale audio datasets and can learn generalized acoustic embeddings that capture rich musical and semantic information from raw audio signals. Models such as MERT provide powerful feature representations that can be transferred to downstream tasks including music genre classification. Attention mechanisms and Bidirectional Gated Recurrent Unit (BiGRU) architectures further improve these representations by learning temporal dependencies and emphasizing important audio patterns during classification. Compared to conventional feature engineering methods, pretrained representation learning approaches provide improved generalization capability and stronger feature discrimination.

This work presents a comparative study of multiple machine learning and deep learning approaches for music genre classification using the GTZAN dataset. The study evaluates traditional machine learning models trained on handcrafted features extracted from both 30-second and 3-second audio clips. Principal Component Analysis (PCA) is applied to analyse the impact of dimensionality reduction on classification performance. Furthermore, a spectrogram-based 2D CNN architecture is developed for deep feature learning from spectrogram images. The study also investigates pretrained audio representation learning using MERT embeddings combined with Attention and BiGRU architectures for downstream genre prediction.

The main objective of this research is to analyse the effectiveness of different audio representations and classification techniques for automated music genre recognition. Comparative evaluation is performed using classification accuracy, macro F1-score, and confusion matrices to study the strengths and limitations of each approach. The experimental findings demonstrate that pretrained embedding-based deep learning models provide superior classification performance compared to conventional handcrafted feature-based methods by learning more discriminative and context-aware audio representations.

II. RELATED WORKS

Music genre classification has attracted significant research attention in the field of Music Information Retrieval due to its importance in recommendation systems, playlist organization, and automatic music retrieval. Earlier approaches primarily relied on handcrafted feature extraction techniques combined with traditional machine learning classifiers. Over the years, researchers have explored deep learning architectures, recurrent neural networks, attention mechanisms, spectrogram-based analysis, and pretrained representations to improve classification performance and robustness.

Yang Yu et al. proposed a deep attention-based music genre classification framework using Bidirectional Recurrent Neural Networks (BRNNs), GRU units, and attention mechanisms for spectrogram analysis. The authors argued that different temporal regions of an audio spectrogram do not contribute equally to genre prediction. To address this limitation, they introduced serial and parallelized attention mechanisms capable of focusing on important temporal segments within spectrogram representations. Their CNN-based parallel attention model demonstrated improved classification performance compared to conventional CNN architectures. However, the approach requires higher computational complexity due to the integration of recurrent and attention-based modules.

Mitt Shah et al. presented a comparative study of machine learning and deep learning approaches for music genre classification using the GTZAN dataset. The authors utilized handcrafted audio descriptors such as spectral centroid, spectral contrast, zero-crossing rate, spectral bandwidth, spectral flatness, and MFCC features for training machine learning models including Support Vector Machines (SVM), Random Forest, and XGBoost classifiers. In addition, Mel-spectrogram images were used as input to a Convolutional Neural Network (CNN) architecture for deep learning-based classification. Their study concluded that CNN-based approaches outperform traditional machine learning methods due to their ability to automatically learn discriminative features from spectrogram representations. However, the deep learning model required larger computational resources and longer training time.

Ndiatenda Ndou et al. conducted a review and comparative analysis of deep learning and traditional machine learning techniques for music genre classification. The study explored multiple categories of audio features including magnitude-based, tempo-based, pitch-based, and chordal progression features. The authors evaluated both 30-second and 3-second audio representations using machine learning classifiers and CNN-based models. Their analysis demonstrated that shorter 3-second segmented audio clips can

improve classification accuracy by increasing the number of training samples. The study also highlighted the effectiveness of MFCC aggregation and spectral descriptors in representing audio characteristics. Despite achieving improved performance, the handcrafted feature extraction process still depended heavily on manual feature engineering and selection techniques.

Pulkit Dwivedi et al. introduced a Generative Adversarial Network (GAN)-based framework for music genre classification. Their approach combined GANs with Fourier Transform and Wavelet Transform techniques to extract both frequency-domain and time-domain audio representations. The generator network produced synthetic audio samples to augment the training data, while the discriminator learned genre-specific audio characteristics. The study demonstrated that GAN-based augmentation improves classification robustness and reduces overfitting by increasing dataset diversity. Although the method achieved high classification accuracy, the training process of GANs remained computationally intensive and unstable in certain scenarios.

Chun Pui Tang et al. proposed a hierarchical Long Short-Term Memory (LSTM)-based architecture for music genre classification. The study explored sequential modeling of MFCC features using LSTM networks and introduced a divide-and-conquer hierarchical classification strategy. Music genres were initially separated into strong and mild categories before further classification into individual genres. Their hierarchical approach improved classification accuracy compared to traditional single-network architectures by capturing temporal dependencies more effectively. However, the model performance decreased as the number of genres increased, indicating limitations in handling highly overlapping musical characteristics.

The previous studies demonstrate that both traditional machine learning and deep learning approaches contribute significantly to music genre classification. Handcrafted feature-based methods provide efficient and interpretable representations but often suffer from limited generalization capability. Deep learning models such as CNNs, LSTMs, GANs, and attention-based architectures improve classification performance by learning complex temporal and spectral patterns directly from audio representations. However, many existing approaches focus only on a single representation method or classification strategy. This research aims to provide a comprehensive comparative analysis of handcrafted feature-based classification, dimensionality reduction using PCA, spectrogram-based CNN learning, and pretrained audio representation models using the GTZAN dataset. The study further evaluates the effectiveness of Attention and BiGRU-based architectures for improving genre prediction performance across multiple audio representations.

III. PROPOSED METHODOLOGY

The proposed framework performs a comprehensive comparative analysis of multiple machine learning and deep learning approaches for music genre classification using the GTZAN dataset. The framework is designed around three major audio representation strategies: handcrafted statistical audio features, spectrogram image analysis, and pretrained audio representation learning from original waveform signals. Each approach processes the same dataset using different

feature extraction and classification methodologies to analyze their effectiveness in identifying music genres.

The GTZAN dataset consists of 1000 audio tracks categorized into 10 genres, namely blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. Each audio sample has a duration of 30 seconds with a sampling rate of 22050 Hz. The dataset was processed into multiple representations to enable fair comparative evaluation across traditional machine learning and deep learning models.

A. Handcrafted Feature-Based Classification

The first stage of the proposed framework focuses on traditional machine learning using handcrafted statistical audio descriptors extracted directly from the waveform signals. In this approach, the Librosa audio processing library was utilized to extract multiple low-level spectral, temporal, and harmonic audio features capable of representing the acoustic characteristics of different music genres.

Initially, two different feature datasets were utilized. The first dataset contains one feature vector extracted from the complete 30-second music clip, while the second dataset divides every 30-second audio track into multiple smaller 3-second segments. The segmentation process significantly increases the number of training samples and enables the classifiers to learn localized temporal musical patterns more efficiently. The 3-second segmented dataset also improves model generalization by exposing the classifier to multiple acoustic variations within the same music track.

Several handcrafted audio descriptors were extracted from every audio signal including Mel Frequency Cepstral Coefficients (MFCCs), chroma frequencies, spectral centroid, spectral bandwidth, spectral contrast, spectral roll-off, zero-crossing rate, Root Mean Square (RMS) energy, and tempo-related descriptors. MFCCs were extracted to model the timbral characteristics of the music signal and represent the short-term spectral envelope of the audio. Chroma features were used to capture harmonic and pitch-related information by representing the distribution of the twelve pitch classes. Spectral centroid and spectral bandwidth were extracted to analyze the brightness and frequency spread of the signal, while spectral contrast and spectral roll-off were used to capture differences between spectral peaks and frequency distribution boundaries. Similarly, zero-crossing rate was used to model rhythmic and percussive characteristics of the audio waveform.

For every extracted feature sequence, statistical aggregation operations such as mean and variance were computed to transform variable-length temporal descriptors into fixed-length numerical feature vectors suitable for machine learning classification. After feature aggregation, the handcrafted feature representation initially contained 58 numerical dimensions for both the 30-second and 3-second datasets. The extracted features were then normalized using StandardScaler to ensure uniform feature distribution and prevent features with larger numerical ranges from dominating the training process.

The genre labels were encoded using LabelEncoder and the normalized feature vectors were divided into training and testing subsets. Multiple machine learning classifiers were trained and evaluated including Logistic Regression, Support Vector Machine (SVM), Random Forest Classifier, and Gradient Boosting-based models. These classifiers were

selected to compare linear learning, nonlinear decision boundaries, ensemble-based learning, and boosting strategies for music genre recognition. Since the 3-second segmented dataset generates multiple predictions for a single music track, a majority voting mechanism was implemented during inference. In this process, all segment-level predictions corresponding to the same music file were aggregated and the most frequently predicted genre was selected as the final output class. This strategy improved classification robustness by reducing the impact of noisy or ambiguous segments.

To further improve computational efficiency and eliminate redundant feature information, Principal Component Analysis (PCA) was applied before classifier training. PCA transforms the original high-dimensional feature space into a lower-dimensional orthogonal representation while preserving the maximum variance information from the dataset. Before applying PCA, the handcrafted feature vectors contained 58 dimensions. After applying PCA with 95% variance preservation, the dimensionality of the 30-second dataset was reduced from 58 dimensions to 33 principal components, while the 3-second segmented dataset was reduced from 58 dimensions to 39 principal components. The reduced feature space significantly lowered computational complexity, minimized feature correlation, reduced overfitting, and improved model generalization while retaining the majority of the discriminative audio information required for genre classification.

B. Spectrogram Image-Based Deep Learning

The second stage of the proposed framework focuses on spectrogram image-based deep learning for music genre classification. Unlike traditional handcrafted feature extraction methods, this approach enables the neural network to automatically learn hierarchical spectral and temporal audio representations directly from visual representations of music signals. Initially, the raw waveform audio signals were converted into Mel-spectrogram images using Short-Time Fourier Transform (STFT) followed by Mel-scale frequency conversion. STFT divides the audio waveform into overlapping temporal windows and computes the frequency spectrum for every segment, thereby preserving both time-domain and frequency-domain information simultaneously. The Mel-scale transformation converts the frequency axis into a perceptually meaningful representation based on human auditory perception. The generated spectrogram images visually represent the energy distribution of frequencies across time, where the horizontal axis represents time, the vertical axis represents frequency, and pixel intensity represents signal amplitude.

After spectrogram generation, the images were resized and normalized before being provided as input to the deep learning architecture. Instead of designing a custom CNN from scratch, the proposed framework utilizes the pretrained ResNet18 architecture for feature extraction and classification. ResNet18 was selected because of its superior feature learning capability and its ability to overcome the vanishing gradient problem through residual skip connections. Comparative experimental analysis showed that ResNet18 achieved better classification performance and improved convergence stability compared to conventional CNN architectures.

The ResNet18 architecture consists of 18 deep convolutional layers organized into multiple residual learning

blocks. The initial convolution layer extracts low-level spectral features such as harmonic transitions, rhythm boundaries, and frequency edges from the spectrogram images. As the network depth increases, the residual blocks progressively learn higher-level musical representations including harmonic textures, rhythmic structures, and genre-specific spectral distributions.

The residual skip connections within ResNet18 allow the network to preserve information from earlier layers and prevent degradation problems commonly observed in deeper convolutional architectures. This enables the model to learn more discriminative and robust audio representations without suffering from gradient instability during training. Since the architecture was pretrained on ImageNet, transfer learning was utilized to initialize the network weights with pretrained feature representations before fine-tuning the model on the spectrogram dataset. The final fully connected classification layer of ResNet18 was modified to perform ten-class music genre classification corresponding to the GTZAN dataset categories. Dropout regularization was incorporated during training to reduce overfitting and improve generalization capability. The model was trained using categorical cross-entropy loss and optimized using the Adam optimizer with mini-batch gradient descent.

During training, the earlier convolutional layers of ResNet18 learned low-level spectrogram patterns while the deeper residual blocks captured complex temporal-frequency relationships associated with different genres. The network automatically learned discriminative audio representations directly from spectrogram images without requiring manual feature engineering. Compared to handcrafted statistical descriptors, the ResNet18-based spectrogram framework demonstrated stronger capability in capturing complex harmonic structures, temporal transitions, and spectral texture variations present across different music genres.

C. Pretrained Audio Representation Learning Using Original Audio Signals

The third stage of the proposed framework focuses on pretrained audio representation learning using the original waveform audio signals. Instead of relying on manually engineered features or spectrogram image generation, this approach utilizes transformer-based pretrained music representation models capable of learning generalized semantic embeddings directly from raw waveform inputs.

The pretrained model utilized in this framework is the Music Embedding Representation Transformer (MERT-v1-95M). The model was loaded using the HuggingFace Transformers library through the AutoProcessor and AutoModel interfaces. Before feature extraction, all audio signals were resampled from 22050 Hz to 24000 Hz to satisfy the input requirements of the pretrained transformer encoder.

The MERT model processes the raw waveform audio signal and generates contextual hidden embedding representations represented by the tensor dimensions (B, T, D), where B represents batch size, T corresponds to temporal sequence length, and D denotes embedding dimension. These transformer embeddings capture rich acoustic, harmonic, temporal, and semantic information learned during large-scale self-supervised pretraining on music datasets.

To convert the variable-length embedding sequences into fixed-dimensional vectors suitable for downstream

classification, both mean pooling and max pooling operations were applied independently across the temporal dimension. Each pooling operation generated a 768-dimensional representation. The pooled representations were then concatenated together to form a final 1536-dimensional audio embedding vector for every music sample. This combined representation preserves both dominant embedding activations and overall contextual semantic information extracted from the audio signal.

The extracted embedding vectors were subsequently reshaped into sequential format and passed into an Attention-enhanced Bidirectional Gated Recurrent Unit (BiGRU) architecture for temporal dependency learning. The BiGRU network was configured with an input dimension of 1536 and hidden size of 256. Since bidirectional processing was enabled, the output feature representation became 512-dimensional. The Bidirectional GRU processes the embedding sequences in both forward and backward temporal directions, enabling the network to learn long-range contextual musical dependencies more effectively.

To further improve temporal feature discrimination, an attention mechanism was incorporated after the BiGRU layer. The attention module consists of two fully connected layers where the 512-dimensional BiGRU output is projected into a 128-dimensional intermediate representation followed by Tanh activation. A second linear transformation converts the intermediate features into scalar attention scores. Softmax normalization was then applied across the temporal dimension to generate normalized attention weights.

The attention mechanism enables the model to identify important temporal embedding regions contributing strongly toward genre prediction. Instead of treating all temporal regions equally, the network dynamically assigns higher importance to discriminative musical segments such as rhythmic transitions, harmonic variations, or instrumental changes. The weighted attention context vector was further combined with max-context pooling to preserve both contextual and dominant temporal embedding information. The resulting feature representation was passed through dropout regularization with a dropout rate of 0.3 followed by a fully connected classification layer that maps the learned embedding representation into ten output genre classes. The final classification probabilities were generated using softmax activation.

In addition to pretrained embedding learning, synthetic audio augmentation experiments were also performed using Generative Adversarial Networks (GANs) to improve dataset diversity and reduce overfitting during training. The GAN architecture consists of a generator network responsible for synthesizing artificial audio representations and a discriminator network based on ResNet18 which simultaneously performs real-versus-synthetic discrimination and genre classification. The discriminator learns more robust genre representations by distinguishing between generated and real audio patterns, thereby improving overall model generalization capability.

Compared to traditional handcrafted feature extraction and spectrogram-based CNN learning, the pretrained MERT representation framework demonstrated significantly stronger semantic understanding of music signals because the embeddings capture contextual musical relationships learned from large-scale audio corpora. The integration of transformer

embeddings, Bidirectional GRU temporal learning, and attention mechanisms enables the framework to learn highly discriminative and context-aware representations for automated music genre classification.

IV. RESULTS AND DISCUSSION

The effectiveness of the proposed music genre classification framework was evaluated using multiple machine learning and deep learning approaches on the GTZAN dataset. The experiments were conducted to analyse the impact of handcrafted statistical audio features, dimensionality reduction using PCA, spectrogram image-based deep learning using ResNet18, and pretrained transformer-based audio representation learning using MERT embeddings. Performance evaluation was carried out using classification accuracy and macro F1-score to ensure balanced evaluation across all ten music genres. The dataset was separated into training and testing subsets prior to model training to ensure unbiased evaluation. For the handcrafted feature-based approaches, experiments were conducted separately on both the 30-second dataset and the 3-second segmented dataset. The 3-second segmentation strategy increased the number of training samples and enabled the classifiers to learn localized temporal audio patterns more effectively. Similarly, for deep learning and pretrained representation learning approaches, all models were evaluated on unseen test samples to analyse real-world generalization capability.

A. Handcrafted Feature-Based Classification

The handcrafted feature-based classification experiments were conducted using statistical audio descriptors extracted from the GTZAN dataset using Librosa. Initially, the extracted feature vectors contained 58 dimensions representing spectral, temporal, harmonic, and energy-related audio characteristics. Multiple machine learning classifiers including Support Vector Machine (SVM), Random Forest, Logistic Regression, and Gradient Boosting were evaluated on both the 30-second and 3-second segmented datasets. For the 30-second dataset, the Random Forest classifier achieved an accuracy of 77.5% with a macro F1-score of 0.7786, while the SVM achieved 74.0% accuracy with a macro F1-score of 0.7344. The 3-second segmented dataset initially produced lower segment-level accuracy due to limited contextual information within shorter clips.

However, after applying majority voting and mean probability aggregation, the classification performance improved significantly. The SVM classifier achieved the best handcrafted feature-based performance with 79.0% accuracy and a macro F1-score of 0.7812 using majority voting. To further reduce feature redundancy and computational complexity, Principal Component Analysis (PCA) was applied with 95% variance preservation. The original 58-dimensional feature space was reduced to 33 dimensions for the 30-second dataset and 39 dimensions for the 3-second dataset. After PCA transformation, the 3-second SVM model achieved 80.0% accuracy with a macro F1-score of 0.7925 using mean aggregation. Experimental analysis showed that PCA improved generalization by removing correlated features while preserving most of the discriminative audio information required for genre classification.

B. Spectrogram Image-Based Deep Learning

The spectrogram image-based deep learning experiments were performed using Mel-spectrogram images generated from raw waveform audio signals using Short-Time Fourier Transform (STFT) and Mel-scale frequency conversion. Initially, conventional CNN architectures were evaluated and achieved an accuracy of 66.5% with a macro F1-score of 0.6606. However, experimental analysis revealed that the pretrained ResNet18 architecture significantly outperformed the baseline CNN due to its residual skip connections and stronger hierarchical feature extraction capability. The ResNet18-based framework achieved an accuracy of 79.5% with a macro F1-score of 0.7968. The earlier layers of ResNet18 captured low-level harmonic and frequency transitions, while the deeper residual blocks learned high-level rhythmic structures and genre-specific temporal-frequency relationships. Transfer learning using ImageNet pretrained weights further improved convergence stability and feature learning efficiency. Compared to handcrafted statistical

descriptors, the ResNet18-based spectrogram framework demonstrated stronger capability in distinguishing visually similar genres by learning more discriminative spectral representations directly from spectrogram images.

C. Results of Pretrained Audio Representation Learning

The pretrained audio representation learning framework was evaluated using transformer-based MERT embeddings

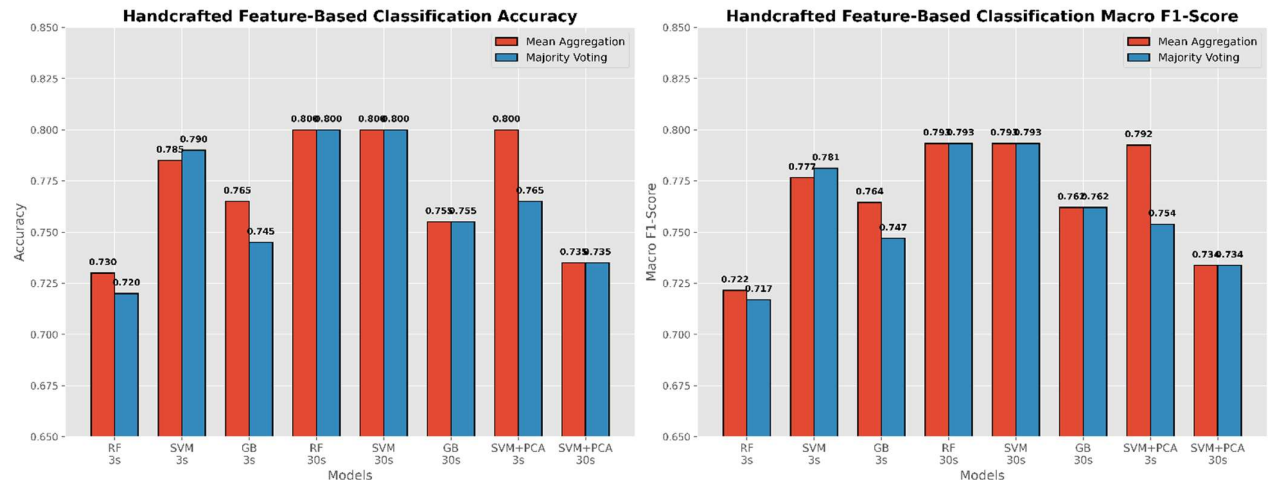


Fig. 1: Comparative Performance Analysis of Accuracy and Macro F1-Score for Handcrafted Feature-Based Classification Models

extracted directly from raw waveform audio signals. Initially, the pretrained embeddings were provided to a Multilayer Perceptron (MLP) classifier, which achieved an accuracy of 81.5% with a macro F1-score of 0.8129. To improve temporal dependency learning, the extracted embeddings were further processed using an Attention-enhanced Bidirectional GRU architecture. The Attention + BiGRU model achieved an accuracy of 76.33% with a macro F1-score of 0.7577. Finally, the complete integrated framework combining MERT embeddings, Attention mechanisms, and Bidirectional GRU achieved the best overall performance with an accuracy of 86.5% and a macro F1-score of 0.8644. The superior performance of the proposed framework can be attributed to the ability of pretrained transformer embeddings to capture deep semantic, harmonic, and contextual musical relationships learned from large-scale audio datasets during self-supervised pretraining. In addition, synthetic audio augmentation experiments using GAN-based approaches achieved accuracies in the range of 73%–74%, improving dataset diversity but still remaining lower than the transformer-based representation learning framework. Overall, the experimental results demonstrate that pretrained contextual audio representation learning significantly outperforms handcrafted feature engineering and spectrogram-based deep learning approaches for automated music genre classification.

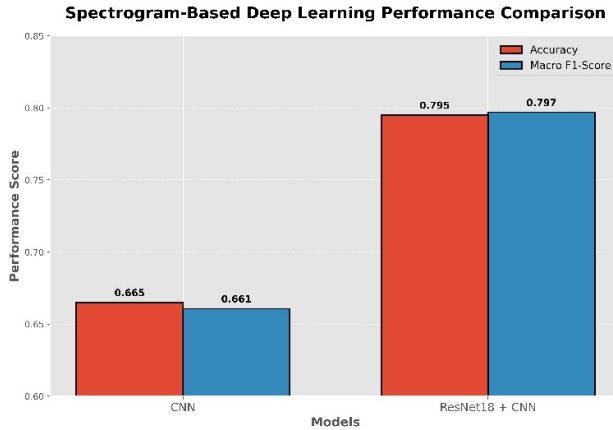


Fig. 2: Comparative Performance Analysis of Accuracy and Macro F1-Score for Spectrogram-Based Deep Learning Models

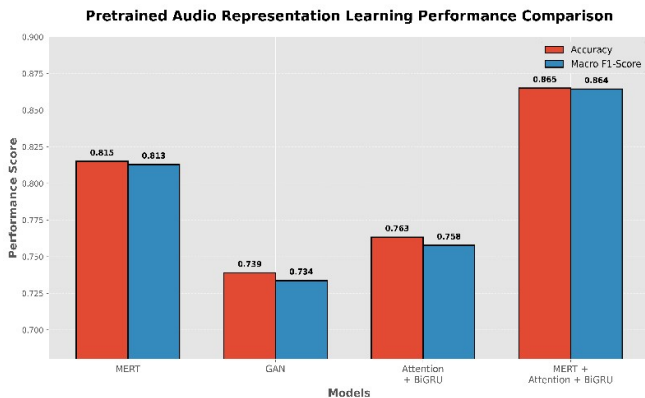


Fig. 3: Comparative Performance Analysis of Accuracy and Macro F1-Score for Pretrained Audio Representation Learning Models

V. CONCLUSION

This study presented a comparative analysis of multiple machine learning and deep learning approaches for music genre classification using the GTZAN dataset. The research evaluated handcrafted statistical audio features, PCA-based dimensionality reduction, spectrogram image-based deep learning using ResNet18, and pretrained transformer-based audio representation learning using MERT embeddings with Attention and Bidirectional GRU architectures. Experimental results showed that handcrafted feature-based classifiers provide reliable baseline performance, while PCA helps reduce feature redundancy and computational complexity without significant information loss.

The ResNet18-based spectrogram framework achieved improved classification performance by learning hierarchical spectral representations directly from Mel-spectrogram images. However, the pretrained MERT-based framework achieved the best overall performance with the highest accuracy and macro F1-score by capturing rich semantic and contextual musical relationships from raw waveform audio signals. Overall, the study demonstrates that pretrained transformer-based audio representation learning significantly outperforms traditional handcrafted feature engineering and conventional deep learning methods for automated music genre classification. Future work can focus on larger datasets, hybrid transformer architectures, and advanced augmentation techniques to further improve classification robustness and generalization capability.

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